

Ruminants	Monogastrics
Rumen, R, omasum, abomasum	Proventriculus, ventriculus
Digestion by microflora	Digestion by enzymes
NDF $\rightarrow$ 30-35%	NDF $<$ 5%
Glucose and VFA production	Absent
MCP	Absent.

In Rumen:

$CH_2O \rightarrow$ saccharides.

~~Starch~~ Cellulose (structural carbohydrate)
 $\downarrow$ cellulase
 Glucose (Non-structural carbohydrate)
 + VFA (Acetic, propionic, butyric acid)

MCP.

$\rightarrow$ Microbial crude Protein.
 $\hookrightarrow$ High quality protein.

* Cereal grains $\rightarrow$ Carbohydrates

* Maize (corn) $\rightarrow$ high percentage for poultry.

Digestive system

* Very less testabds. ($<$ 300) (1000 in human, 1700 in pep.)

① Beak, Pharynx

② Oesophagus

③ crop.

④ Proventriculus

⑤ Ventriculus

→ Esophagus is muscular organ.

↳ Take feed to stomach.

→ Grop. is temp. feed storage

→ Proventriculus is glandular stomach

↳ Have paired glands which secrete enzymes such as pepsin etc.

→ Ventriculus / Mechanical stomach / Gizzard

↳ Grinding of feed

↳ Two types of muscles.

↳ Yellow membrane on muscles called kiolin ~~added~~ to protect muscles from acidic secretions.

↳ Colour due to bile.

→ Intestine:

→ Small intestine.

↳ Duodenum.

↳ Duodenal loop.

↳ Connect Pancreas.

↳ Low enzyme secretion.

↳ Jejunum.

↳ Most absorption and digestion (Villi)

↳ Ileum. (less absorption than jejunum)

→ Large intestine.

→ Caeca

↳ 2 blind pouches.

→ Colon

→ Rectum.

→ Cloaca.

↳ Common opening of 3 systems

↳ digestive, urinary, reproductive.

Feed:

↳ Moisture.

↳ Dry matter.

↳ Nutrients

Feed:

Roughages

→ High fiber > 80%.

→ Low TDN < 60%.

→ Low in energy

Concentrates

Low fiber

→ High TDN

→ High in energy.

NDF:

Neutral detergent fiber

∝ $\perp$
intake

ADF:

Acid detergent fiber

∝ $\perp$
Digestibility

* Fiber → cellulose, hemicellulose, Pectin